

Camden Conrad

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Software Engineer and Systems Programmer building toward flight software. Author of Besom, an open-source deterministic simulation harness for NASA's core Flight System (cFS), and Cauldron, an IDE enforcing NASA/JPL coding standards as live diagnostics. Shipped commercial software (Axis Football 2026, 10K+ players) with a 7.8× production tool speedup, and daily-drives a self-built Linux environment. Specialized in Rust, determinism, zero-GC design, and the discipline of software that assumes a bit flipped somewhere.

TECHNICAL SKILLS

Languages: Rust · C · C++ · C# · Java · TypeScript · JavaScript · Python · SQL · WGSL · HLSL

Graphics & Rendering: wgpu · Custom Renderers · Custom Shader Pipelines · Unreal Engine 5 · Unity · Emissive & Lighting Systems

Performance & Systems: Zero-GC · Manual Memory Management · SIMD · Multithreading · ECS · Data-Oriented Design · Thread Safety · Perf Profiling · Wayland/Smithay · sched_ext · Deterministic Simulation · Debugging & Instrumentation

Tools & Other: NASA cFS/OSAL · NASA/JPL Coding Standards · ILGPU · Avalonia UI · WFC / Proc-Gen · Cloudflare · WebSockets · Editor Tooling

PROFESSIONAL EXPERIENCE

Junior Software Engineer · Farrell Holdings Inc · Remote

May 2026 – Present

- NDA — Internal full stack development (JavaScript/TypeScript, Rust, Python, SQL).

Contract Systems Programmer · Axis Games · Remote · Axis Football 2026

April 2025 – Present

- Shipped Axis Football 2026 on PC & console (10,000+ players) as credited Contract Systems Programmer.
- Rebuilt zero-GC Replay System from scratch — 25 MB for 15s of full 22-player data with variable-speed playback; added instrumentation that eliminated all timing-related crashes in QA.
- Re-architected Crowd Editor pipeline — 684% (7.8×) speedup, cutting designer iteration from days to minutes while supporting 10× more crowd volume.
- Automated live statistics content pipeline — parses real-time player/season data into structured narrative commentary at scale.

FEATURED PROJECTS

Besom — Deterministic Simulation Harness for NASA cFS — Rust · NASA cFS/OSAL · Apache-2.0 · github.com/camdenconrad/besom

- Open-source harness that runs real, unmodified cFE flight software but owns the clock — time advances only per granted tick, so identical scenarios produce byte-identical telemetry, making flight software truly regression-testable (a gap in NASA's real-clock NOS3 simulator).
- Propagates a two-body orbit in lockstep with the simulation and doubles as a ground station — pause the vehicle exactly where the software stopped.

Cauldron — GPU-Rendered Rust IDE for Flight-Software Discipline — Rust · wgpu · Static Analysis · Custom LSP/Debugger Clients

- Enforces NASA/JPL coding standards (Power of Ten, JPL C Standard) as live save-time diagnostics, including whole-project call-graph analysis such as recursion detection that a language server alone cannot express; replicates cFS CI checks locally.

RuneOS — Custom Linux Environment (Daily Driver) — Rust · Wayland/Smithay · Custom Kernel · sched_ext

- Built and daily-drive a full custom Linux stack: packaged kernel (linux-rune), Wayland compositor on Smithay, native Rust shell, package manager, ~30 system daemons, and a native app suite including the Cauldron IDE — with portions held to the JPL coding standard.

Bellatrix — Isometric Face Composition Editor — Rust · wgpu · WGSL · winit · Engine-less

- Custom face-based voxel editor and engine-less renderer built from scratch — face primitives with ghost placement, depth testing, 5 selection modes, full undo/redo; proprietary format stores 20K+ faces at ~2 MB.

repliKate — High-Velocity Learning System — C# · Custom ML · ILGPU · GPU Acceleration

- Lightweight continuous learning engine — adapts without pretraining at ~100 MB RAM. Thread-safe online learning pipeline applied to Chess AI (MCTS), TDM game AI (A* pathfinding, bullet-dodge), and ERA5/NASA POWER weather prediction.

EDUCATION

University of North Carolina at Charlotte

2021 – 2026

Computer Science (no degree) — completed core CS curriculum through senior project (100%) · Concentration: AI, Robotics & Game Development